Tutorial 9 report
EEE5311 (Digital IC design)

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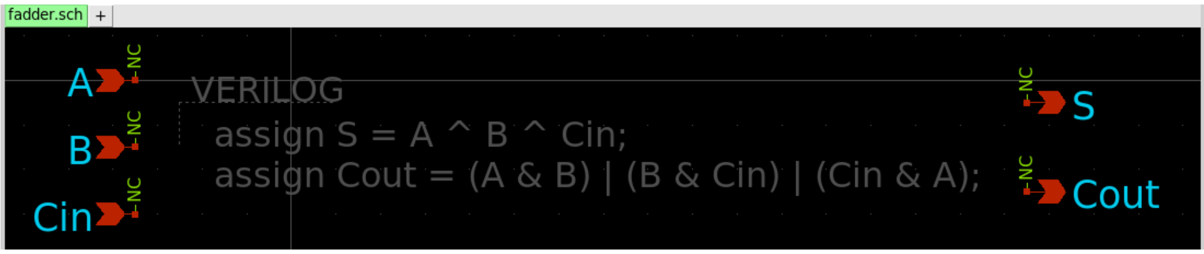
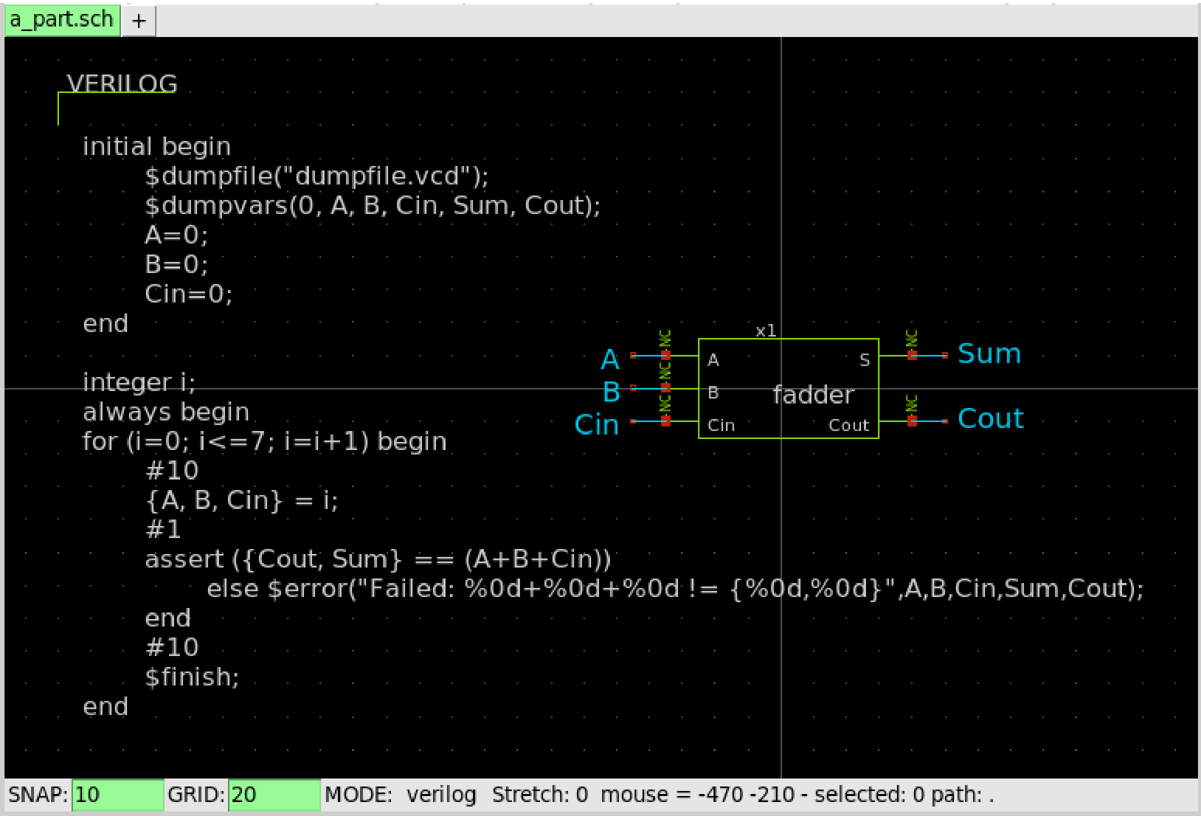
Question 1

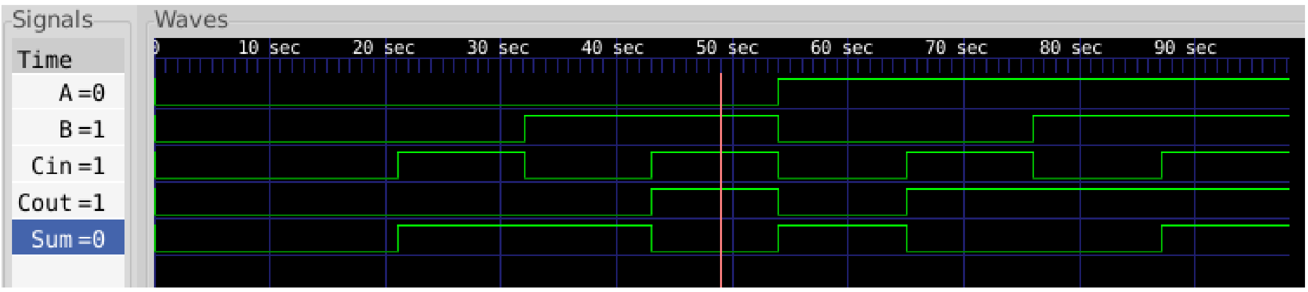
- Verilog schematic for full adder –



- Testbench containing the full adder symbol –

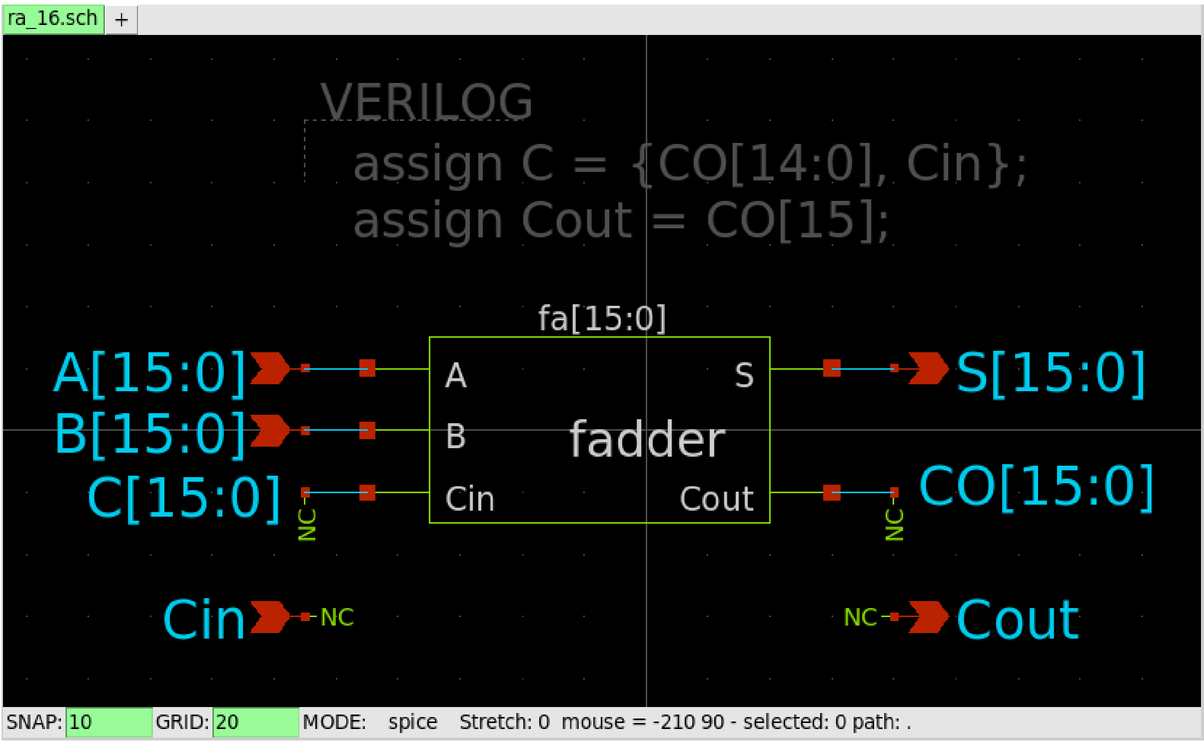


- Gtkwave output showing the output signals for all possible input combinations –

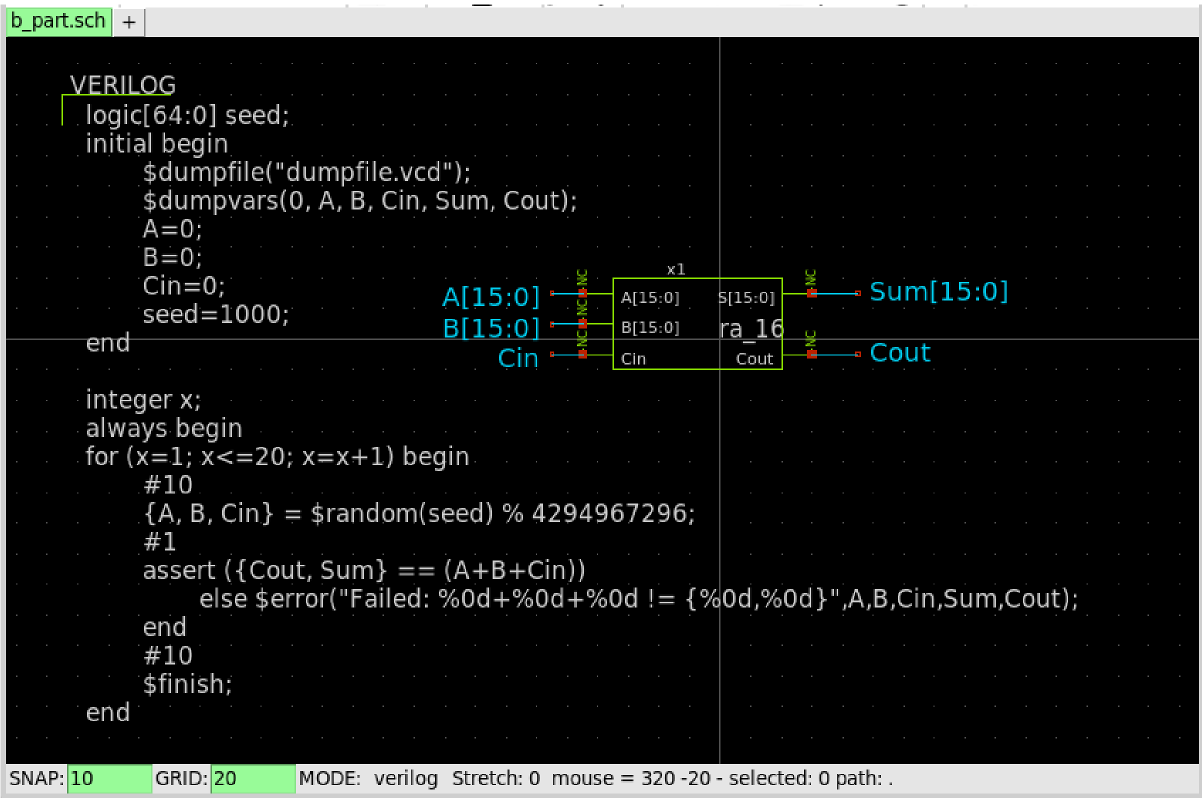


Question 2

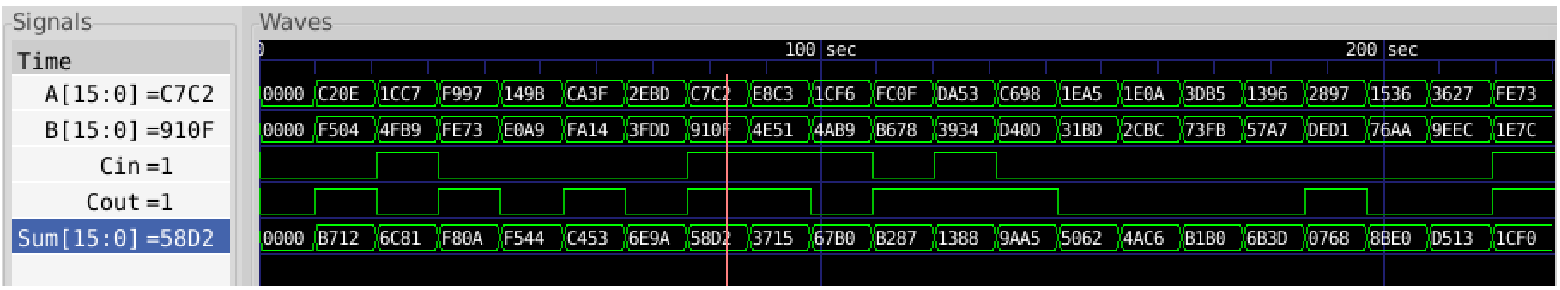
- The Verilog (xschem) schematic for the 16-bit full adder used –



- The testbench used to test it for 20 random inputs –

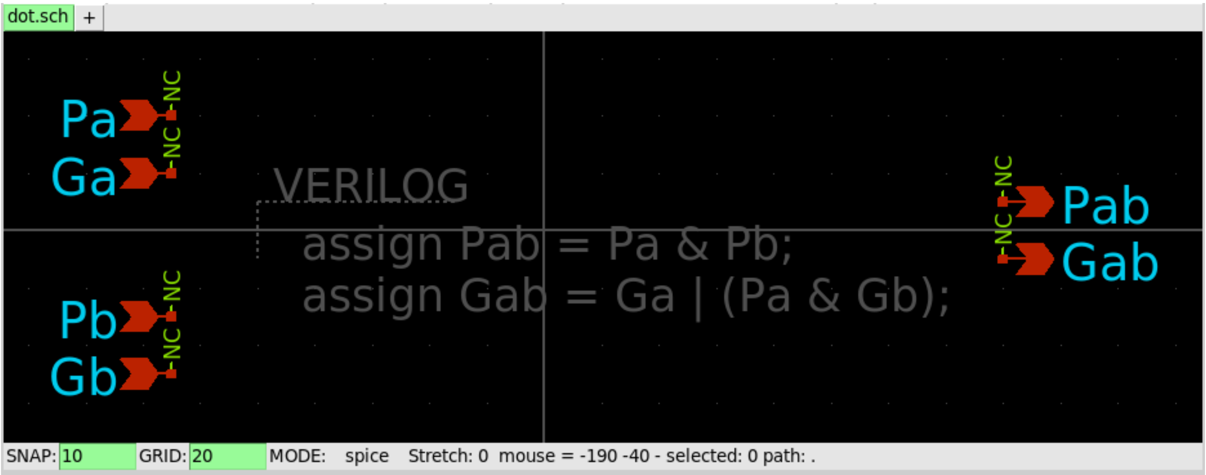


- Gtkwave output for the 20 random inputs –

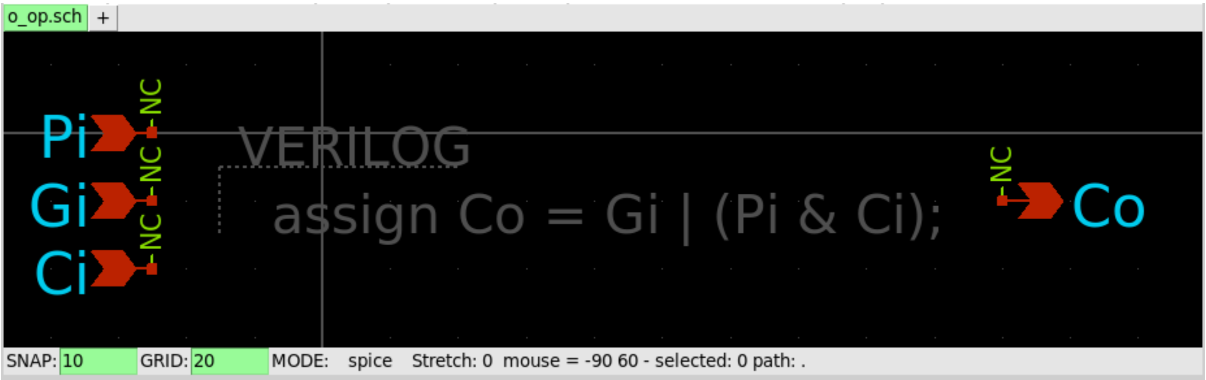


Question 3

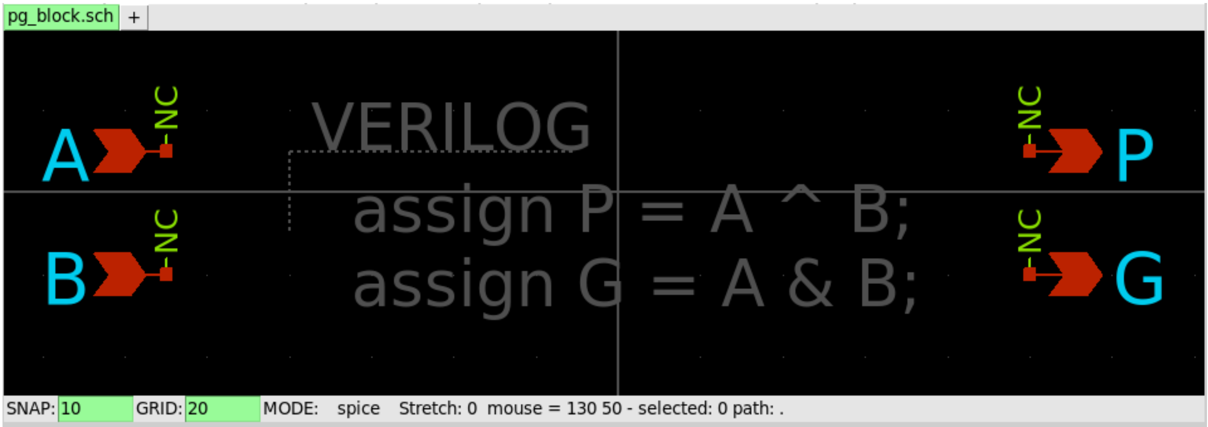
- Since the Verilog code used in the testbenches for the Brent-Kung and Kogge-Stone adders was the same as the 16-bit ripple carry adder in the 2nd question (including the random generator seed value), the testbenches and gtkwave outputs for the former adders have not been included.
- Blocks used that are common to both adders –
  - The dot function (Propagate/Generate module) –



  - The circle function (special case of dot function) –

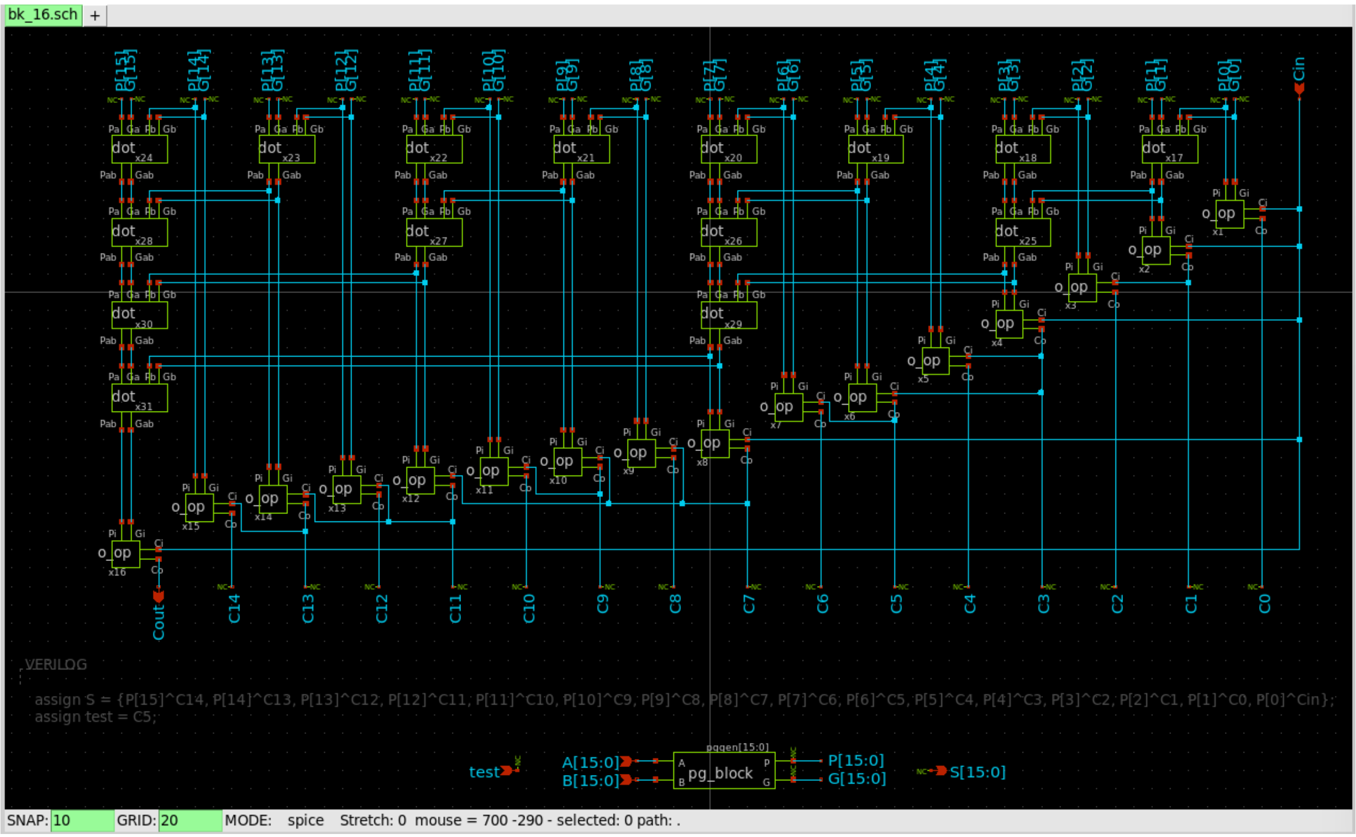


  - Block for generating propagate (P) and generate (G) signals given inputs A and B –



Part a)

- The Brent-Kung adder's Verilog implementation in an xschem schematic –



Part b)

- The Kogge-Stone adder's Verilog implementation in an xschem schematic –